

Window functions

Running Total

```
SELECT duration_seconds,  
       SUM(duration_seconds) OVER (ORDER BY start_time) AS running_total  
FROM tutorial.dc_bikeshare_q1_2012
```

```
SELECT start_terminal,  
       duration_seconds,  
       SUM(duration_seconds) OVER  
         (PARTITION BY start_terminal ORDER BY start_time)  
         AS running_total  
FROM tutorial.dc_bikeshare_q1_2012  
WHERE start_time < '2012-01-08'
```

The usual SUM, COUNT, and AVG

```
SELECT start_terminal,  
       duration_seconds,  
       SUM(duration_seconds) OVER  
         (PARTITION BY start_terminal) AS running_total,  
       COUNT(duration_seconds) OVER  
         (PARTITION BY start_terminal) AS running_count,  
       AVG(duration_seconds) OVER  
         (PARTITION BY start_terminal) AS running_avg  
FROM tutorial.dc_bikeshare_q1_2012  
WHERE start_time < '2012-01-08'
```

ROW_NUMBER()

```
SELECT start_terminal,  
       start_time,  
       duration_seconds,  
       ROW_NUMBER() OVER (ORDER BY start_time)  
          AS row_number  
FROM tutorial.dc_bikeshare_q1_2012  
WHERE start_time < '2012-01-08'
```

```
SELECT start_terminal,  
       start_time,  
       duration_seconds,  
       ROW_NUMBER() OVER (PARTITION BY start_terminal  
          ORDER BY start_time)  
          AS row_number  
FROM tutorial.dc_bikeshare_q1_2012  
WHERE start_time < '2012-01-08'
```

RANK() and DENSE_RANK()

```
SELECT start_terminal,  
       duration_seconds,  
       RANK() OVER (PARTITION BY start_terminal  
                   ORDER BY start_time)  
       AS rank  
FROM tutorial.dc_bikeshare_q1_2012  
WHERE start_time < '2012-01-08'
```

NTILE

```
SELECT start_terminal,  
       duration_seconds,  
       NTILE(4) OVER  
       (PARTITION BY start_terminal ORDER BY duration_seconds)  
       AS quartile,  
       NTILE(5) OVER  
       (PARTITION BY start_terminal ORDER BY duration_seconds)  
       AS quintile,  
       NTILE(100) OVER  
       (PARTITION BY start_terminal ORDER BY duration_seconds)  
       AS percentile  
FROM tutorial.dc_bikeshare_q1_2012  
WHERE start_time < '2012-01-08'  
ORDER BY start_terminal, duration_seconds
```